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Operando Spectroscopic Analysis of Single-Sn-Atom Material for Electrochemical CO₂ Reduction

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25 **Abstract**

26 Sn-based materials have been demonstrated as promising catalysts for selective electrochemical CO₂
27 reduction reaction (CO₂RR). However, the detailed structures of catalytic intermediates and the key
28 surface species remain to be identified. In this work, a series of single-Sn-atom catalysts with well-
29 defined structures are developed as model systems to explore their electrochemical reactivity towards
30 CO₂RR. The selectivity and activity of CO₂ reduction to formic acid on Sn single-atom sites are shown
31 to be correlated with Sn(IV)-N₄ moieties axially coordinated with oxygen (O-Sn-N₄). Employing a
32 combination of *operando* X-ray absorption spectroscopy, attenuated total reflectance surface enhanced
33 infrared absorption spectroscopy, Raman spectroscopy and ¹¹⁹Sn Mössbauer spectroscopy, surface-
34 bound bidentate tin carbonate species are precisely captured during CO₂RR. Moreover, the electronic
35 and coordination structure of the single-Sn-atom species under reaction conditions are accurately
36 determined. Density functional theory (DFT) calculations further supports the preferred formation of
37 Sn-O-CO₂ species over the O-Sn-N₄ sites, which effectively modulates the adsorption configuration of
38 the reactive intermediates and lowers the energy barrier for the hydrogenation of *OCHO species, as
39 compared to the preferred formation of *COOH species over the Sn-N₄ sites, thereby greatly
40 facilitating CO₂-to-HCOOH conversion.

41

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Introduction

The overdependence on fossil fuels leads to an excessive accumulation of greenhouse gas in the atmosphere. It is urgent to develop highly efficient and environmentally friendly technologies to mitigate the depletion of fossil resources, abate carbon emission and to switch to renewable resources.¹⁻

⁵ Electrocatalytic carbon dioxide reduction reactions (CO₂RRs) offer sustainable opportunities to transform CO₂ into value-added chemicals and carbon-based fuels.⁶⁻⁸ While various products can be produced from CO₂RR, some of the C1 products (i.e. formic acid and CO) can be obtained with high selectivity and high energy efficiency, and are of great interest to the chemical industry.⁹⁻¹² Formic acid (HCOOH) is widely explored as a hydrogen storage material and chemical fuel for fuel cells.¹³⁻¹⁵ Until now, electrocatalysts developed for reducing CO₂ to HCOOH have included noble metals (e.g., Pd) and transitional-metal-based materials (e.g., Cu, Bi, Pb, In, Sn etc).¹⁶⁻²²

Owing to their unique electronic structure, Sn-based materials have been identified as attractive catalysts for electrochemical CO₂ reduction to HCOOH and/or CO,²³ but suffer from insufficient selectivity and activity, especially at large cathodic current densities.²⁴ To this end, past researches focused on doping Sn crystalline phases with other metal or nonmetal elements to tune the electron density at the Sn active sites and their neighboring dopants to selectively stabilize the reaction intermediates and improve the activity and selectivity of CO₂RR.^{19, 25-29} However, the complex structure model and the complicated coordination environment of metal sites make it difficult to capture the intermediate species and determine the true active sites during the reaction process.

Single-atom catalysts (SACs) with well-defined and atomically-dispersed metal sites, tunable coordination environment and high utilization efficiency, have shown promising CO₂RR activity and selectivity towards CO, especially Fe-N-C and Ni-N-C materials.³⁰⁻³¹ This class of materials provides an ideal platform for exploring the structure of active sites and understanding structure-activity relationships.³²⁻³⁷ Beyond 3d transition metal SACs, early studies reported that single-Sn-atom sites in

Sn-SACs could selectively reduce CO₂ either to HCOOH or CO.^{2, 10 38} To reveal the intrinsic origin for the different CO₂RR selectivity of Sn-single-atom sites, it is necessary to pinpoint the exact electronic and coordination structure of the catalytic species under CO₂RR conditions. This in turn requires the rational design of single-Sn-atom materials with well-defined sites and the development of advanced *operando* characterization tools with atomic resolution.

In this work, a series of Sn-N-C catalysts featuring single-Sn-atoms with tunable and well-defined coordination environments were developed. The average Sn(IV)/Sn(II) ratio could be modulated via the functionalization of carbon nanotubes, modifying the adsorption energy of Sn phthalocyanine (SnPc) molecules. In particular, a model single-Sn-atom catalyst with ~100% axially oxygen coordinated Sn(IV) species (O-Sn-N₄) was obtained using OH-functionalized carbon nanotubes, showing the optimal catalytic activity and selectivity towards HCOOH. Taking advantages of *operando* attenuated total reflectance surface enhanced infrared absorption spectroscopy (ATR-SEIRAS) and Raman spectroscopy, a surface-bound bidentate tin carbonate species was revealed as a key reaction intermediate during CO₂RR. *Operando* ¹¹⁹Sn Mössbauer spectroscopy was applied for the first time to follow changes in the electronic and coordination structure of single-Sn-atom species under CO₂RR. In combination with *operando* X-ray absorption spectroscopy (XAS), the *in situ* generation of the Sn-O-CO₂ catalytic intermediate during CO₂RR was clearly identified. Density functional theory (DFT) calculations further evidenced the dynamic evolution of the electronic structure of O-Sn-N₄ site for CO₂RR and the catalytic mechanism over single-Sn-atom species was unveiled at atomic scale.

Results and discussion

Characterization of the single-Sn-atom catalysts

A series of single-Sn-atom catalysts were prepared by immobilizing SnPc molecules on CNTs subjected to different functionalization treatments (CNT, CNT-OH and CNT-NH₂) via π - π

interaction.³⁹ The respective electrocatalysts are labelled SnPc/CNT, SnPc/CNT-OH and SnPc/CNT-NH₂. The synthesis of SnPc/CNT-OH is schemed in **Fig. 1a**, and fully detailed in **Methods** and **Supplementary Fig. 1**. The X-ray diffraction (XRD) patterns of this series of materials show only one diffraction peak assignable to the diffraction line of carbon nanotubes, suggesting that the SnPc molecules are well-dispersed and therefore cannot contribute to the diffraction patterns (**Supplementary Fig. 2**). In contrast, Raman spectra display signals from both molecular SnPc and CNTs for all materials (**Supplementary Fig. 3**), indicating that SnPc was successfully loaded onto the CNTs. The Sn contents were in the range of 0.37-0.66 wt. % (**Supplementary Table 1**). **Figure 1b** and **Supplementary Fig. 4** show representative scanning electron microscope (SEM) and transmission electron microscope (TEM) images of the catalysts, on which no aggregation of metal species could be observed. The surface morphology of SnPc/CNT-OH is similar to that of CNT-OH, indicating uniform dispersion of SnPc on CNT-OH, which is also confirmed by energy-dispersive X-ray spectroscopy (EDX) mapping analysis (**Supplementary Fig. 5**). Furthermore, high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image of SnPc/CNT-OH in **Fig. 1c** clearly shows Sn atoms dispersed on CNT and the absence of Sn clusters or nanoparticles, further suggesting the successful formation of single-Sn-atom catalysts.

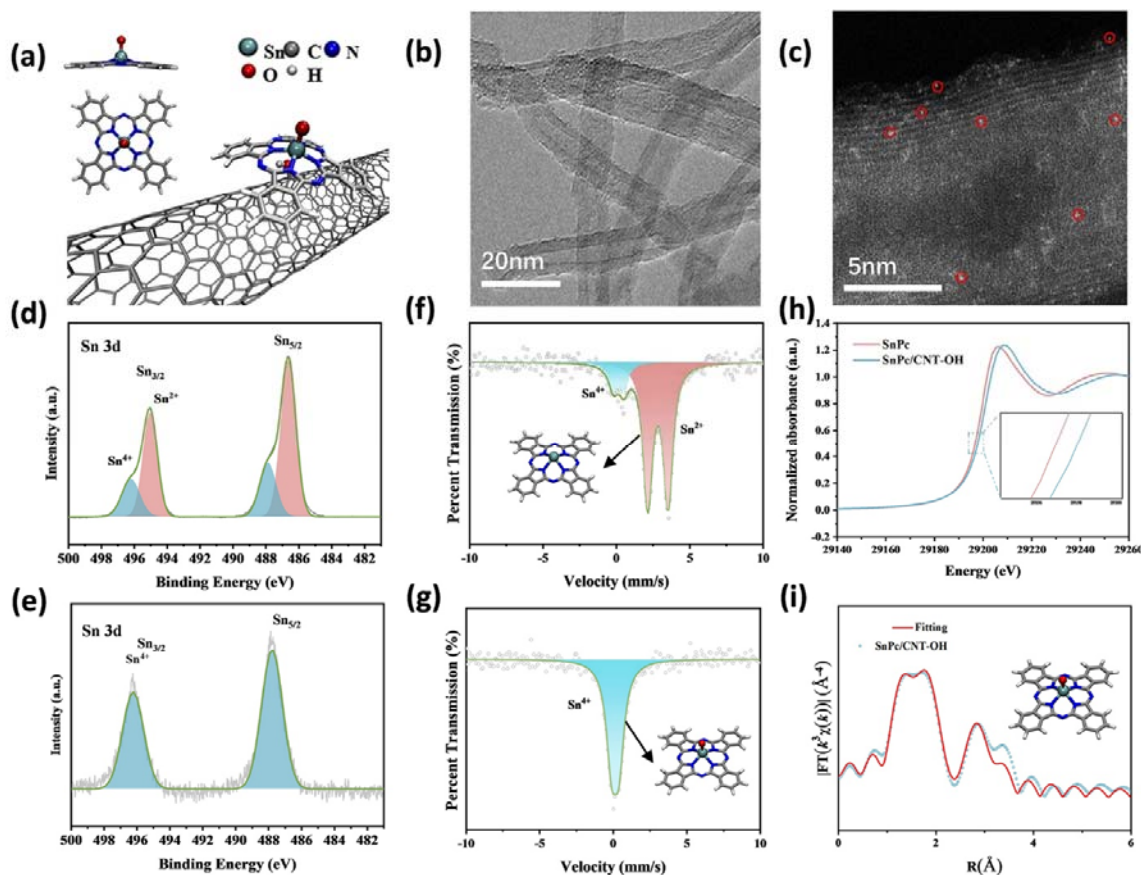
X-ray photoelectron spectroscopy (XPS) was performed to examine the elemental compositions and Sn valence states of the single-Sn-atom catalysts. The Sn 3d XPS spectrum of the commercial SnPc can be deconvoluted into two peak groups assigned to Sn(II) (486.6 eV, 495 eV) and Sn(IV) (488.5 eV, 496.4 eV) (**Fig. 1d** and **Supplementary Fig. 6**).⁴⁰ For the series of catalysts, the Sn(IV)/Sn(II) ratio increases strongly in the order SnPc < SnPc/CNT-NH₂ < SnPc/CNT < SnPc/CNT-OH (**Supplementary Table 2** and **Supplementary Fig. 6**), with relative Sn(IV) amounts increasing from 18.1 % to 100 %. In particular, only Sn(IV) is observed for SnPc/CNT-OH (**Fig. 1e**), indicating that OH⁻ functional groups on CNT selectively lead to Sn(IV) possibly due to the strength of electrostatic interactions, which is supported by DFT calculations. As compared in **Supplementary**

Fig. 7, CNT-OH exhibits an adsorption energy for Sn(IV) species of -2.379 eV, more negative than those of CNT-NH₂ and CNT (-1.946 and -2.229 eV, respectively), suggesting that the CNT-OH surface more likely adsorbs Sn(IV) species than the other two types of carbon nanotubes.

To further determine the coordination environment and the electronic structure of the single-Sn-atom species, we conducted room temperature ¹¹⁹Sn Mössbauer spectroscopy measurements. As shown in **Fig. 1f**, the commercial SnPc shows two quadrupole doublets with distinct values of isomer shift (IS) (0.03 and 2.72 mm s⁻¹) and quadrupole splitting (QS) (0.70 and 1.39 mm s⁻¹) (**Supplementary Table 3**). The doublets with low and high IS can unambiguously be assigned to Sn(IV) and Sn(II), respectively.⁴¹ Moreover, the IS and QS values for the main high-IS doublet of the SnPc commercial product match well with those reported for Sn(II)Pc.⁴² Interestingly, only one doublet with an IS value of 0.04 mm s⁻¹ and a QS value of 0.50 mm s⁻¹ was observed for SnPc/CNT-OH (**Fig. 1g**), which can be unambiguously assigned to Sn(IV) species. Based on IS and QS values calculated for a set of hypothetical site structures, the fitted parameters of this doublet match well with those calculated by DFT for the axially oxygen coordinated Sn(IV) species (O-Sn-N₄) structure (**Supplementary Fig. 8**).

As displayed in **Fig. 1h**, the absorption edge of Sn K-edge X-ray absorption near edge structure (XANES) spectrum of SnPc/CNT-OH shifts towards higher energies compared to that of SnPc, suggesting a higher average Sn valence state in the SnPc/CNT-OH catalyst,²⁶ in line with XPS analysis. The Fourier transforms of the extended X-ray absorption fine structure (EXAFS) spectra of both SnPc and SnPc/CNT-OH exhibit the dominant Sn-N coordination at 1.5-1.7 Å, distance not corrected for phase shift (**Supplementary Fig. 9**). No obvious peaks related to Sn-Sn coordination could be observed, confirming the atomic dispersion of all or most of the Sn atoms. In addition, the EXAFS spectra for SnPc and SnPc/CNT-OH could be well fitted with the structural models of Sn-N₄ and O-Sn-N₄ (as optimized from DFT), respectively (**Supplementary Fig. 10**). The fitted parameters are summarized in **Supplementary Table 4**. As depicted in **Fig. 1i** and **Supplementary Fig. 10**, the Sn atom in SnPc is coordinated, on average, with four adjacent N atoms (Sn-N₄), whereas the Sn atom in

141 SnPc/CNT-OH, on average, has four neighbouring N atoms and one axially coordinated O atom (O-
 142 Sn-N₄). All these results clearly show the successful development of a model single-Sn-atom catalyst
 143 with 100% axially oxygen coordinated Sn(IV) species anchored on OH-functionalized carbon
 144 nanotubes.



145 **Fig. 1 | Structural characterization of the single-Sn-atom catalysts.** **a**, Schematic illustration for
 146 the preparation of SnPc/CNT-OH. **b**, TEM and **c**, HAADF-STEM image of SnPc/CNT-OH. **d-e**, High-
 147 resolution Sn 3d XPS spectrum for SnPc and SnPc/CNT-OH, respectively. **f-g**, Room-temperature
 148 ¹¹⁹Sn Mössbauer spectra for SnPc and SnPc/CNT-OH, respectively. **h**, Normalized Sn K-edge XANES
 149 spectra for SnPc and SnPc/CNT-OH (inset shows the Sn K-edge XANES spectra in a narrow region
 150 sensitive to the Sn oxidation state). **i**, Experimental and fitted EXAFS spectrum of SnPc/CNT-OH in
 151 *R* space.
 152

Electrochemical CO₂RR performance

The electrochemical CO₂RR performance of SnPc/CNT, SnPc/CNT-OH and SnPc/CNT-NH₂ were examined in an H-type cell filled with CO₂-saturated 0.5 M KHCO₃ (pH = 7.3) electrolyte, and the CO₂RR catalyst loading was fixed at 1 mg·cm⁻².⁴³ The CO₂ reduction activity of the single-Sn-atom catalysts was first investigated by linear scan voltammetry (LSV) measurements. As shown in **Fig. 2a**, the current density for CO₂ reduction over SnPc/CNT-OH is the highest among this series of catalysts. **Fig. 2b** summarizes the potential-dependent CO₂RR Faradaic efficiency for SnPc/CNT, SnPc/CNT-OH and SnPc/CNT-NH₂. The main CO₂RR product is HCOOH, along with a small amount of CO, while H₂ is produced in significant quantities on all three catalysts. The HCOOH Faradaic efficiency over SnPc/CNT-OH can reach 75.4% at -1.13 V vs. RHE, which is higher than that over SnPc/CNT (48%) and SnPc/CNT-NH₂ (18%). The same trend is observed at other potentials, SnPc/CNT-OH systematically leading to higher FE towards HCOOH than the other two catalysts. As displayed in **Supplementary Fig. 11**, the SnPc/CNT-OH exhibits an optimal HCOOH partial current density (j_{HCOOH}) of -14.3 mA·cm⁻² at -1.13 V vs. RHE, higher than that over SnPc/CNT (-5.2 mA·cm⁻²) and SnPc/CNT-NH₂ (-2.4 mA·cm⁻²). Besides excellent activity and selectivity, both the FE towards HCOOH and the current density remained nearly constant over SnPc/CNT-OH during 10 h at -1.13 V vs. RHE, indicating promising stability (**Fig. 2c**). The correlation of Faradaic efficiency, partial current density, and relative contents of Sn(II), Sn(IV) and nitrogen species are shown in **Fig. 2d** and **Supplementary Fig. 12**. The data reveals that the catalytic performance towards formic acid formation is positively correlated with the relative content of Sn(IV) (**Supplementary Fig. 12d**). This suggests that the Sn(IV) single atom species is the dominant catalytic site for electrochemical CO₂ reduction to formic acid, in line with recent reports.⁴⁴

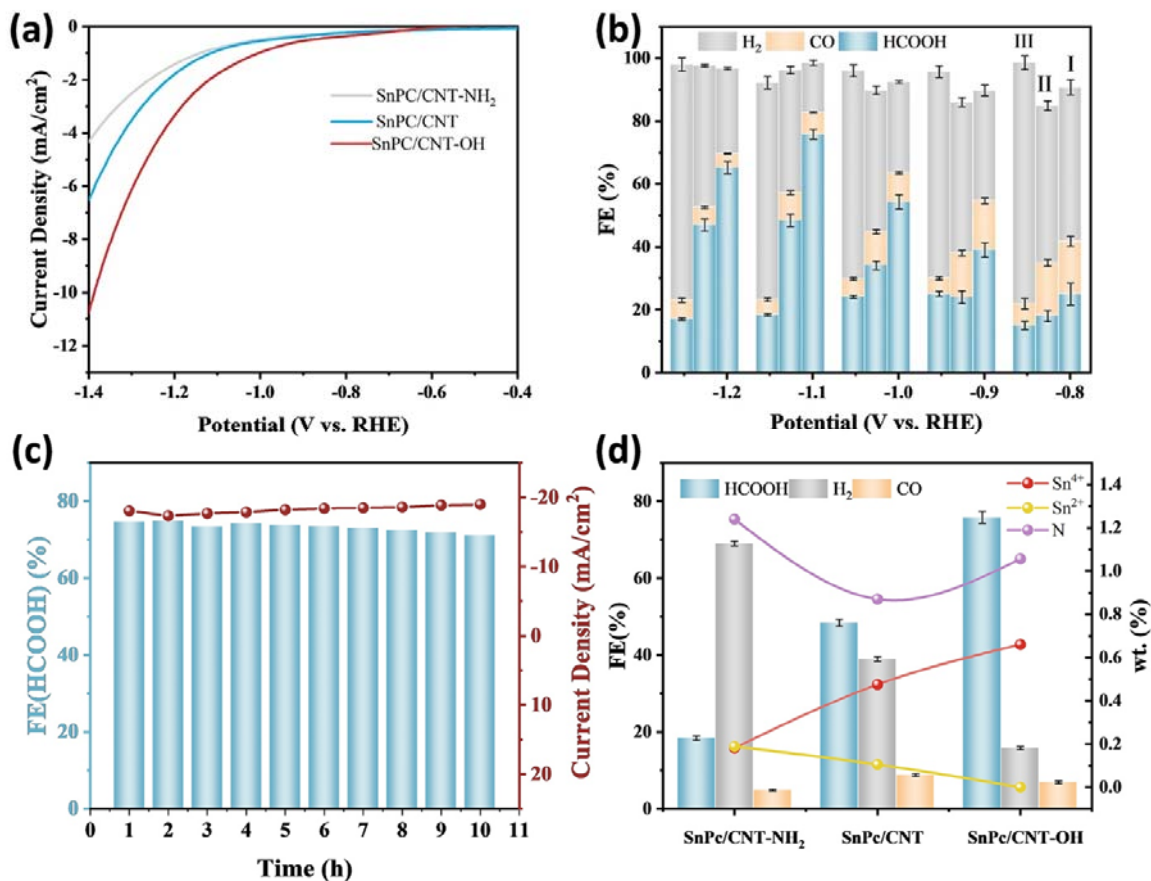


Fig. 2 | Electrochemical CO₂ reduction. **a**, LSV curves acquired on a rotating disc electrode in CO₂-saturated 0.5 M KHCO₃ solution. **b**, Faradaic efficiency of I: SnPc/CNT-OH, II: SnPc/CNT and III: SnPc/CNT-NH₂ at different applied cathodic potentials. **c**, Stability of SnPc/CNT-OH for CO₂RR at -1.13 V vs. RHE. **d**, Correlation diagram of the CO₂RR Faradaic efficiency and the relative contents of Sn(II), Sn(IV) and nitrogen species of SnPc/CNT, SnPc/CNT-OH and SnPc/CNT-NH₂: FE is indicated as columns (left-hand side y axis), while Sn(II), Sn(IV) and N content are indicated as curves (right-hand side y axis).

Identification of the catalytic intermediates and the surface reaction species

XRD, TEM and XPS characterizations of the SnPc/CNT-OH catalyst were performed and compared before and after CO₂RR, to investigate if an irreversible structural evolution occurred (Supplementary Fig. 13-15). Interestingly, no obvious morphology and valence state changes were

observed, further confirming the high stability of the SnPc/CNT-OH catalyst during the CO₂RR process.

To gain insights into the surface reaction intermediates over the single-Sn-atom catalyst during electrochemical CO₂ reduction to HCOOH, *operando* ATR-SEIRAS was carried out.⁴⁵ Control experiments were firstly conducted in Ar-saturated 0.1 M K₂SO₄ solution (**Supplementary Fig. 16**); only one peak centered at 1238 cm⁻¹ was observed, which can be assigned to SO₄²⁻ in the electrolyte. During the potential stepping from 0 to -1.5 V vs. RHE and from -1.5 V vs. RHE to 0 in CO₂-saturated 0.1 M K₂SO₄ solution (pH = 4.5), the intensity of the absorption peak at 1650 cm⁻¹ originating from water vibration was observed to slightly increase when stepping the potential from 0 to -1.5 V vs. RHE, and to decrease for the same potential step but in the opposite direction (**Fig. 3a and b**). The weak bands at 1950 and 2000 cm⁻¹ are related to the linearly bonded CO (CO_L) because of the CO₂ to CO reaction pathway on CNT-OH (**Supplementary Fig. 17**). Additional absorption peaks at 1285 and 1115 cm⁻¹ appeared at -0.8 V vs. RHE and their intensities increased when further decreasing the potential from -0.8 V to -1.5 V vs. RHE. These peaks can be ascribed to the C-O bond vibration in the bidentate tin carbonate.⁴⁶ The intensity of the tin carbonate peaks slowly decreased and finally disappeared when reverting the potential scan from -1.5 to 0 V vs. RHE. All these results clearly evidence the involvement of bidentate tin carbonate intermediates during the CO₂RR over SnPc/CNT-OH.

Operando Raman spectra recorded at varying potentials in CO₂- or Ar-saturated 0.5 M KHCO₃ solution are shown in **Fig. 3c-d** and **Supplementary Fig. 18-19**.⁴⁷ As displayed in **Fig. 3c**, at open circuit potential (OCP), two characteristic Raman peaks at 1342 cm⁻¹ and 1505 cm⁻¹ were observed on SnPc/CNT-OH, which can be assigned to C=C and C-N vibrations of the phthalocyanine ring, respectively.⁴⁸ With potential decreasing from -0.7 to -1.2 V vs. RHE, the peak intensity of C=C and C-N gradually decreased and eventually vanished, which is most probably due to the generation of surface-bound tin-carbonate (bidentate) species with a symmetric structure. Upon switching the

electrode potential back to OCP, the Raman spectrum could almost be restored to that recorded under the initial OCP condition, suggesting that the potential-induced changes for the SnPc/CNT-OH catalyst are reversible. Moreover, *operando* Raman spectra measured in Ar-saturated 0.5 M KHCO₃ solution showed no significant changes in peak intensity, further reflecting that the observed reversible change of characteristic Raman peaks probably resulted from the *in situ* formation of surface-bound bidentate tin carbonate species, which was captured by the *operando* ATR-SEIRAS measurements.

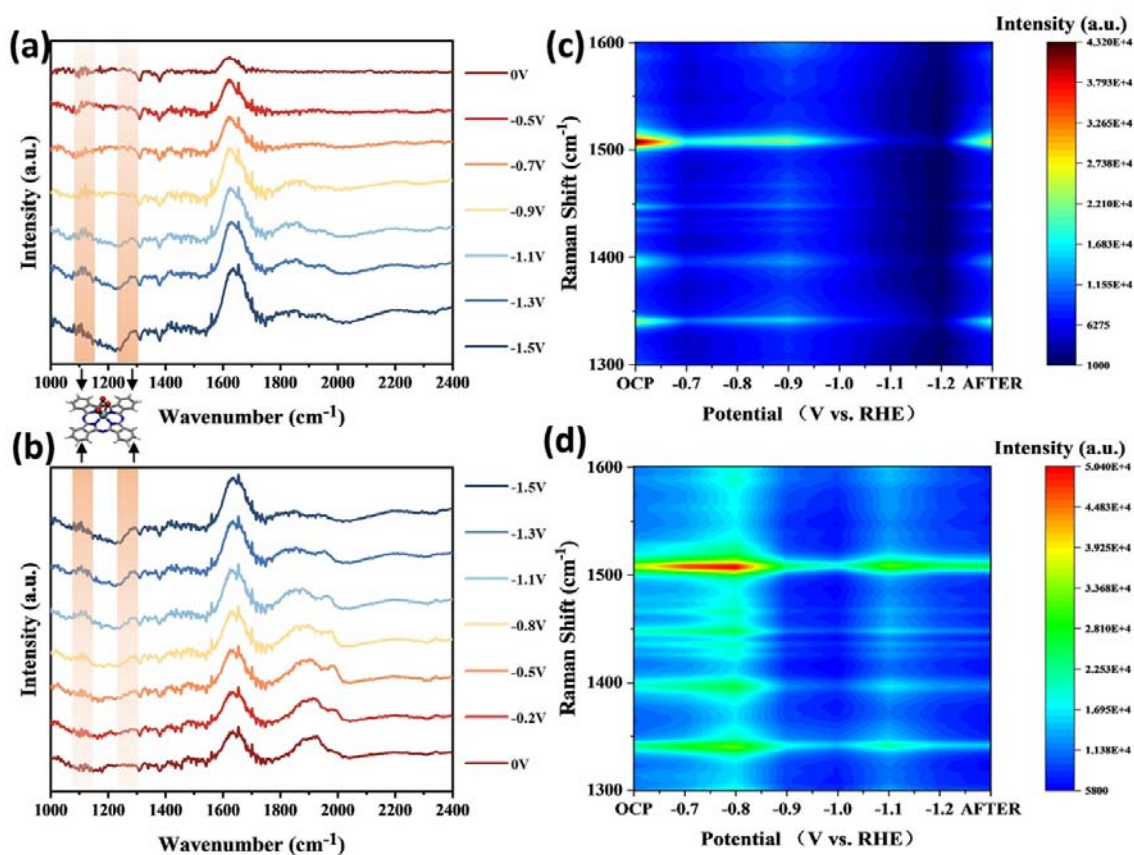
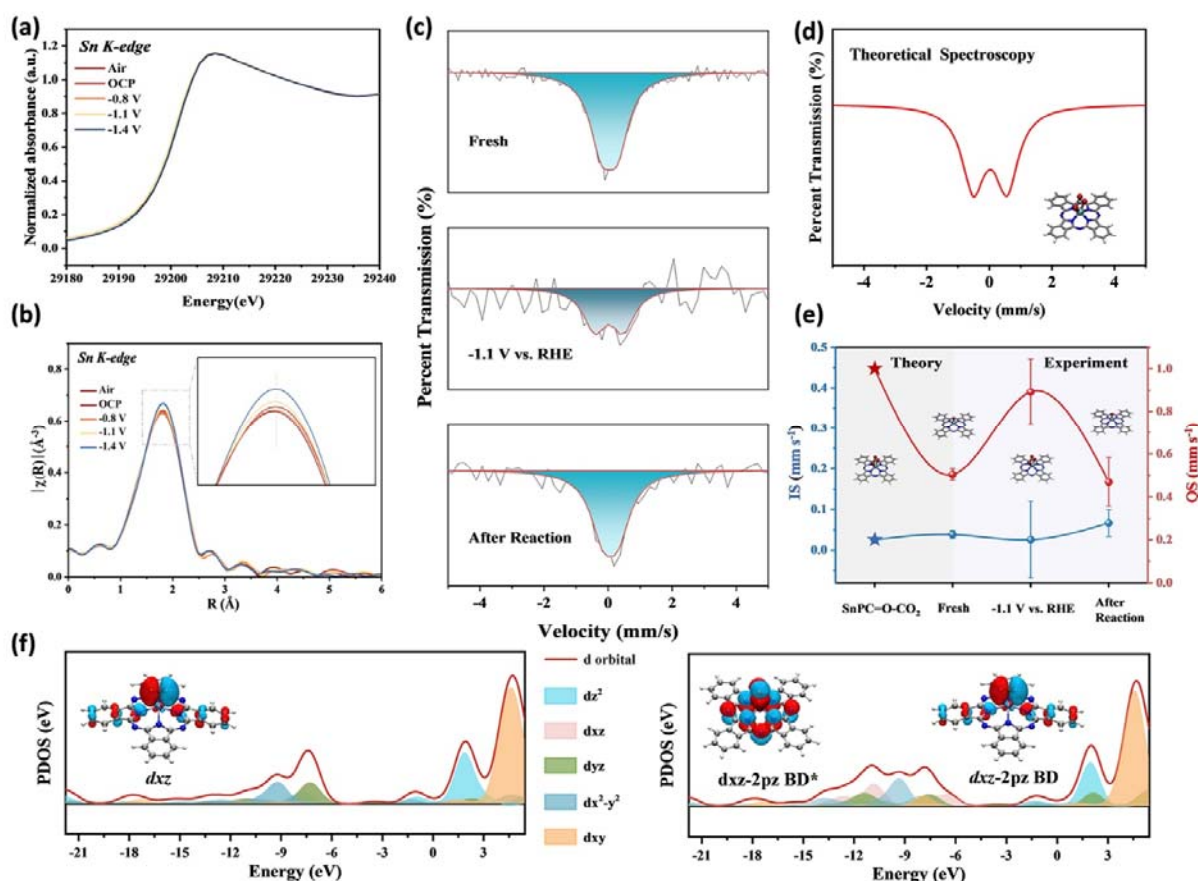


Fig. 3 | Identification of the surface reaction species, a-b, Operando ATR-SEIRAS spectra of SnPc/CNT-OH collected under various applied cathodic potentials (from 0 to -1.5 V vs. RHE and from -1.5 V vs. RHE to 0) in CO₂-saturated 0.1 M K₂SO₄. Operando Raman spectra recorded over SnPc/CNT-OH at various potentials (V vs. RHE) in c, CO₂-saturated and d, Ar-saturated 0.5 M KHCO₃ solution.

224 To further probe the structure of the catalytic intermediates, *operando* XAS measurements were
225 performed. The Sn *K*-edge XANES and EXAFS spectra were recorded in CO₂-saturated 0.5 M KHCO₃
226 solution at various potentials. For the XANES spectra (**Fig. 4a**), a negligible reduction was observed
227 for the Sn *K*-edge of SnPc/CNT-OH during the CO₂RR, revealing that the oxidation state of Sn in
228 SnPc/CNT-OH could be maintained at Sn(IV). In addition, in the EXAFS spectra shown in **Fig. 4b**,
229 the slight increase of the Sn-N (or Sn-C or Sn-O) intensity suggests an increase of the coordination
230 number (CN) of Sn as the CO₂ reduction reaction proceeds. However, negligible changes in the Sn-N
231 bond length were observed, indicating that the coordination environment did not change significantly
232 during the CO₂RR.

233 The electronic and coordination structure of the single-Sn-atom species under CO₂RR conditions
234 ^{47, 49, 50} were further investigated using *operando* ¹¹⁹Sn Mössbauer spectroscopy in a home-made H-
235 type *operando* cell (**Supplementary Fig. 20**). The CO₂-saturated 0.5 M KHCO₃ electrolyte was
236 replaced every ~1.5 h to maintain the pH of the solution. The current-time response of SnPc/CNT-OH
237 during the entire *operando* Mössbauer measurements was quite stable, supporting that the setup
238 performs well and the *operando* spectroscopic data are reliable (**Supplementary Fig. 21**). As shown
239 in **Fig. 4c**, at -1.1 V vs. RHE, the Mössbauer spectrum of SnPc/CNT-OH is different from that of the
240 fresh sample, displaying a symmetric signal with an IS value of 0.03 mm s⁻¹ and a QS value of 0.89
241 mm s⁻¹, matching well with the theoretical Mössbauer spectrum of Sn-O-CO₂ (**Fig. 4d**). The
242 experimental IS and QS values of the catalyst under various *operando* conditions as well as the
243 theoretical IS and QS values of Sn-O-CO₂ are shown in **Fig. 4e** and **Supplementary Table 5**. The
244 results support the *in situ* formation of Sn-O-CO₂ species over O-Sn-N₄ sites during CO₂RR. After
245 removing the applied potential, the Mössbauer spectrum could be fitted by a quadrupole doublet with
246 an IS value of 0.07 mm s⁻¹ and a QS value of 0.50 mm s⁻¹, similar to that of the fresh SnPc/CNT-OH,
247 further suggesting the reversibility of the single-Sn-atom species during the CO₂RR.

248 The dynamic changes observed in the *operando* Mössbauer and XAS measurements could be
 249 further illustrated by the calculated projected density of states (PDOS) of O-Sn-N₄ and Sn-O-CO₂. As
 250 shown in **Supplementary Fig. 22**, similar electron density encloses in the s and p orbital of Sn for O-
 251 Sn-N₄ and Sn-O-CO₂, which results in the negligible changes in the K-edge of *operando* XANES
 252 spectra and IS value of *operando* Mössbauer spectra. However, compared with Sn(IV) in O-Sn-N₄,
 253 the d orbitals (d_{xz} and d_{yz}) of Sn-O-CO₂ change from degeneration to non-degeneration (**Fig. 4f**), which
 254 leads to an increase in the asymmetric distribution of the electrons and thus an increase of QS. All
 255 these results clearly evidenced the *in-situ* formation of Sn-O-CO₂ species over O-Sn-N₄ sites during
 256 the CO₂RR.²⁴



258 **Fig. 4 | Identification of the catalytic intermediates.** **a**, Normalized *operando* Sn K-edge XANES
 259 spectra. **b**, FT magnitude of EXAFS spectra of SnPc/CNT-OH during CO₂RR recorded in CO₂-

saturated 0.5 M KHCO₃ solution. **c**, *Operando* ¹¹⁹Sn Mössbauer spectra for SnPc/CNT-OH recorded before CO₂RR (Fresh), at -1.1 V vs. RHE, and after CO₂RR (After Reaction) in CO₂-saturated 0.5 M KHCO₃ solution at room temperature. **d**; Theoretical ¹¹⁹Sn Mössbauer spectrum of Sn-O-CO₂, **e**, Comparison of the experimental IS and QS values of the catalyst recorded under various conditions and the theoretical IS and QS values of Sn-O-CO₂ structural model (optimized by DFT). **f**, Schematic illustration showing the PDOS of Sn-d orbitals on O-Sn-N₄ and Sn-O-CO₂, respectively.

Theoretical insights into the catalytic mechanism

Theoretical calculations were performed to further investigate the mechanism of CO₂RR over the single-Sn-atom catalysts. The optimized structures of O-Sn-N₄ and Sn-N₄, and the possible adsorption configurations of reaction intermediates are shown in **Supplementary Fig. 23**. **Fig. 5a** shows the Kohn–Sham MO energy-level correlation diagram between CO₂ and the O-Sn-N₄ site during the first CO₂ capture step ($* + \text{CO}_2 \rightarrow *\text{CO}_2$). There exist covalent overlaps between the fully occupied d_{z²} orbital of the O-Sn-N₄ site and the non-occupied π* orbital of CO₂, generating one fully occupied d_{z²}-π* bonding orbital (BD) and one empty d_{z²}-π* antibonding (BD*) orbital. Similarly, the fully occupied d_{xz} orbital of the O-Sn-N₄ site and the 2p_z orbital of CO₂ also generate two fully occupied d_{xz}-2p_z bonding orbital (BD) and d_{xz}-2p_z antibonding (BD*) orbital. The corresponding MO contour plots of these MOs are displayed in **Supplementary Fig. 24**.

We further calculated the Gibbs free energy diagram of CO₂RR to form HCOOH and CO over O-Sn-N₄ and Sn-N₄. As shown in **Fig. 5b**, for the pathway of generating formic acid, the formation of the *HCOOH intermediate via *OCHO/HCOO* hydrogenation is the potential-limiting step; whereas, for producing CO (**Supplementary Fig. 25**), the potential-limiting step is the formation of *COOH. The corresponding energy barriers of CO₂-to-HCOOH and CO₂-to-CO on O-Sn-N₄ and Sn-N₄ are summarized in **Fig. 5c**. Specifically, the downhill energy barrier of the first CO₂-to-*CO₂ step of CO₂RR on O-Sn-N₄ is calculated to be -0.58 eV, while Sn-N₄ exhibits a downhill energy barrier of -

0.13 eV, revealing that the CO₂ activation can take place more readily on O-Sn-N₄. Furthermore, O-Sn-N₄ exhibits a relatively lower energy barrier at the formation of *HCOOH ($\Delta G = 0.8$ eV) than Sn-N₄ ($\Delta G = 0.93$ eV), which shall lead to a higher electrocatalytic activity for CO₂RR to HCOOH over O-Sn-N₄. Additionally, Sn-N₄ displays a much stronger adsorption of *COOH with $\Delta G = 0.78$ eV than O-Sn-N₄ with $\Delta G = 1.02$ eV, resulting in a better catalytic ability for the generation of CO over Sn-N₄. As illustrated in **Supplementary Fig. 26-27**, we also assumed that CO₂ approached the single-Sn-atom sites from the back side of O-Sn-N₄ (O-Sn-N₄-inversion). From the Gibbs free-energy diagram, such a convex inversion shows a much higher energy barrier for the formation of both *COOH and *OCHO intermediates, thereby prohibiting both the HCOOH and CO production. Therefore, the reaction pathways of CO₂RR over O-Sn-N₄ and Sn-N₄ sites are presented in **Fig. 5d** and **Supplementary Fig. 28-29**. The O-Sn-N₄ species can regulate the adsorption configuration of *CO₂, which effectively reduces the energy barrier for the formation of *OCHO intermediate and facilitates the CO₂-to-HCOOH conversion.

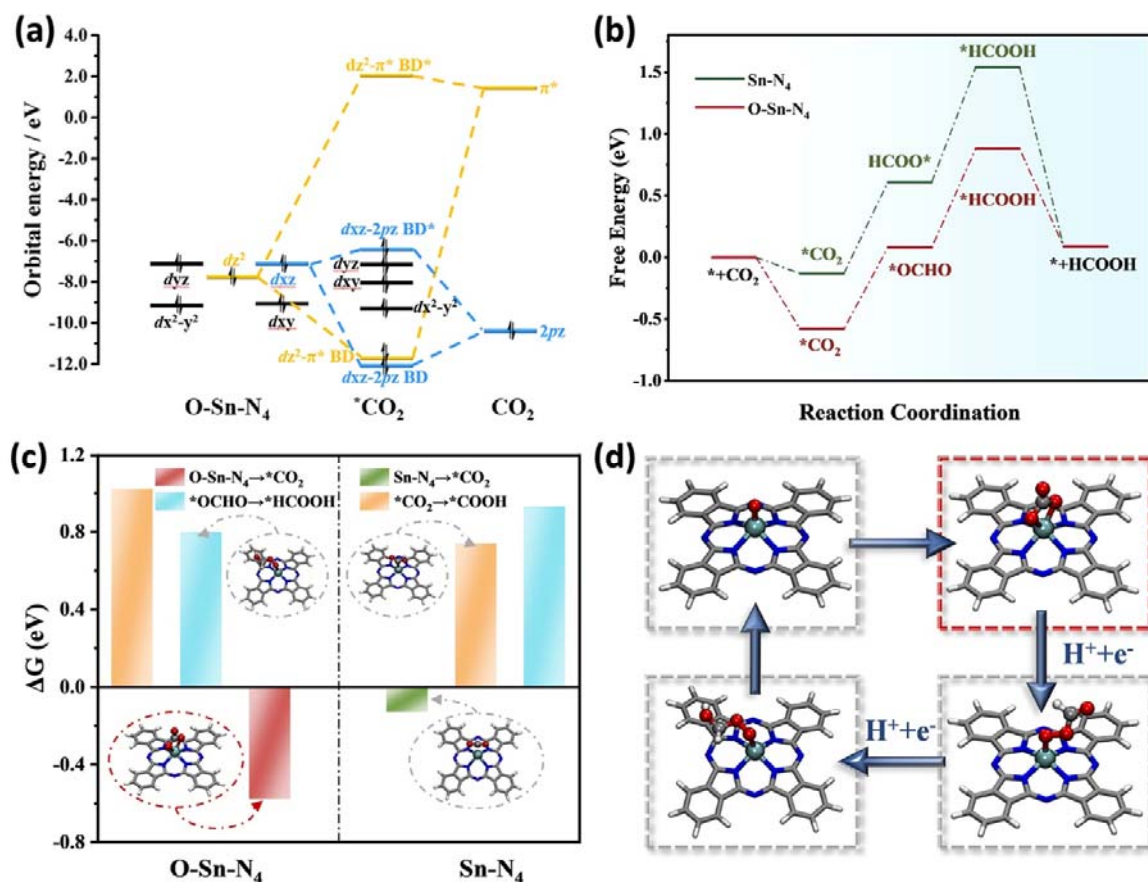


Fig. 5 | Theoretical calculations. **a**, Schematic Kohn–Sham molecular orbital (MO) energy-level correlation diagram between CO₂ and the O-Sn-N₄ site during the CO₂ capture step (* + CO₂ → *CO₂). **b**, Calculated Gibbs free energy diagrams for CO₂RR to HCOOH over O-Sn-N₄ and Sn-N₄ sites. **c**, Calculated ΔG for HCOOH and CO production over O-Sn-N₄ and Sn-N₄ sites. **d**, Proposed reaction pathway for CO₂RR over O-Sn-N₄ site.

Conclusions

In summary, a series of single-Sn-atom catalysts with well-controlled coordination and electronic structure were developed to explore the CO₂ reduction chemistry. Using a combination of *operando* spectroscopies and DFT calculations on a model single-Sn-atom catalyst in which all Sn atoms are in a positive IV oxidation state, situated in a phthalocyanine ring and additionally coordinated to an axial

308 oxygen species (O-Sn-N₄), the *in situ* generation of surface-bound bidentate tin carbonate species (Sn-
309 O-CO₂) in CO₂RR was captured and the dynamic change of the electronic and coordination structures
310 of active sites during CO₂RR was identified. Theoretical studies further revealed that the O-Sn-N₄
311 species could regulate the adsorption configuration of *CO₂ with increasing the asymmetric
312 distribution of the d orbital electrons of Sn(IV), reducing the energy barrier for the formation and
313 hydrogenation of *OCHO species, thus facilitating CO₂-to-HCOOH conversion. The results of this
314 work provide an in-depth understanding of the detailed structures of catalytic intermediates and the
315 key surface species associated with CO₂RR and shall pave the way towards the rational design of high-
316 performance CO₂ electroreduction catalysts.

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Author Contributions

Y.C.D., J.Z. and S.F.W. contributed equally to this work. Y.C.D., X.N.L., Y.Q.H. and B.L. designed the study. Y.C.D. synthesized and characterized the materials. J.Z. performed the density functional theory calculations. Y.C.D. and S.F.W. conducted the electrochemical measurements and analysis. Y.C.D. and S.F.W. performed the Raman spectroscopy measurements and analysis. Y.C.D. and J.D. conducted the ATR-SEIRAS measurements and analysis. H.J.T., W.J.Z., S.F.H. and W.X. performed the XAFS measurements and analysis. Y.C.D., S.F.W. R.R.C., J.H.W. and X.N.L. conducted the Mössbauer spectroscopy measurements and analysis. J.H.W. and F.J. participated in the modification of the Results and Discussion section. Y.C.D., X.N.L. and B.L. wrote and edited the manuscript with inputs from all authors. The project was supervised by X.N.L., Y.Q.H. and B.L.

Notes

The authors declare no competing financial interests.

ASSOCIATED CONTENT

Supporting Information

Supplementary data associated with this article is available free of charge via the Internet at www.nature.com/reprints.

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